Ethanol Blending in Gasoline - Guatemala

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U.S. GRAINS & BIOPRODUCTS COUNCIL

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Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.

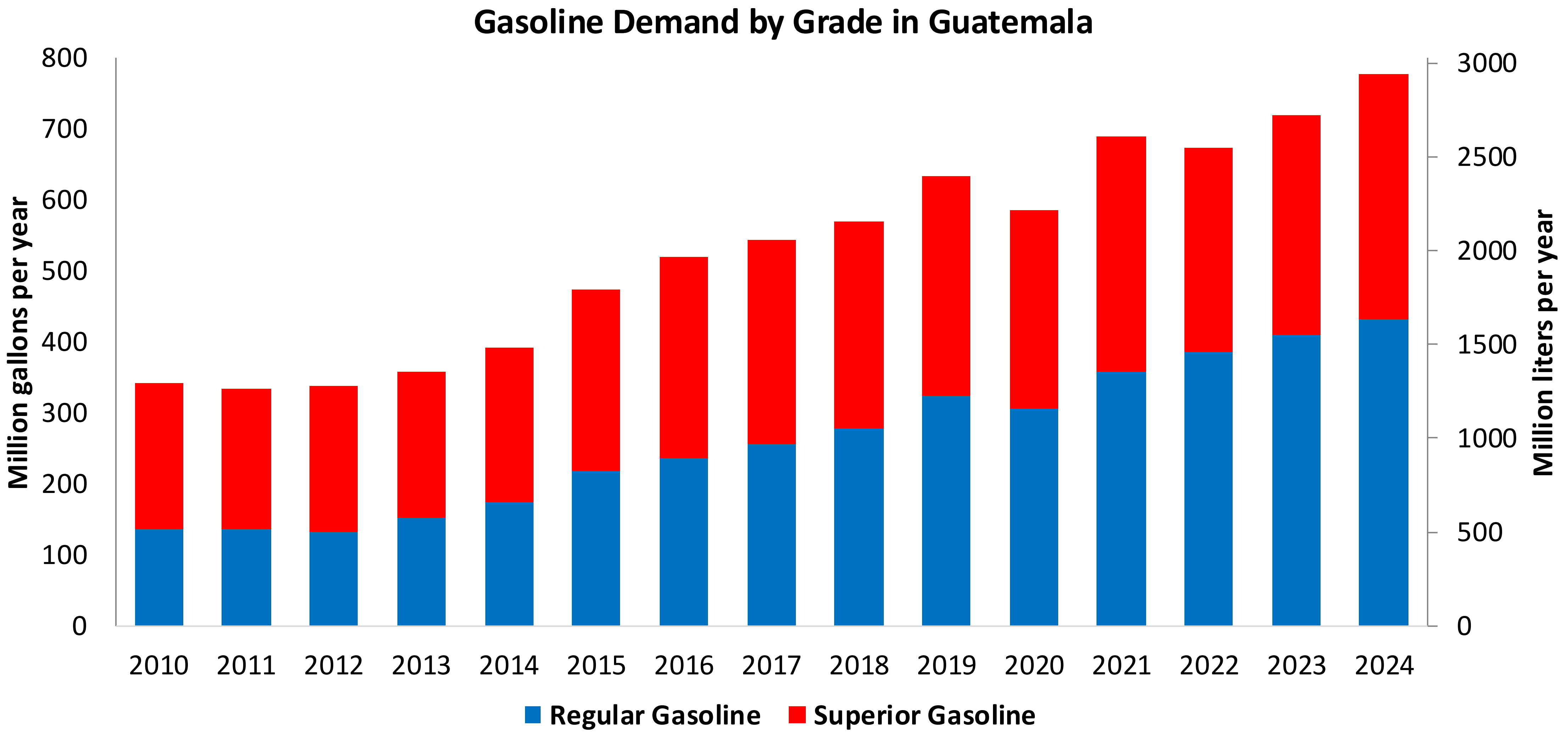
Ethanol Blending in Gasoline - Guatemala

In 2024, gasoline consumption was 778 million gallons (2,946 million liters). Gasoline is supplied only by imports, mainly from United States. There are two grades of gasoline: (RON 91 - AKI 85) and Superior (RON 95 - AKI 89). 55% of gasoline is gasoline Regular.

Guatemala allows for blends up to 10% v/v (Acuerdo Gubernativo 159-2023, MEM) but no ethanol is blended in the country because there is no stakeholder's consensus regarding its implementation. It has local ethanol advanced ethanol production that is exported to Europe. Guatemala is the main ethanol productor in Central America. The E10 mandate is currently under analsys for implementation for 2026-2027.

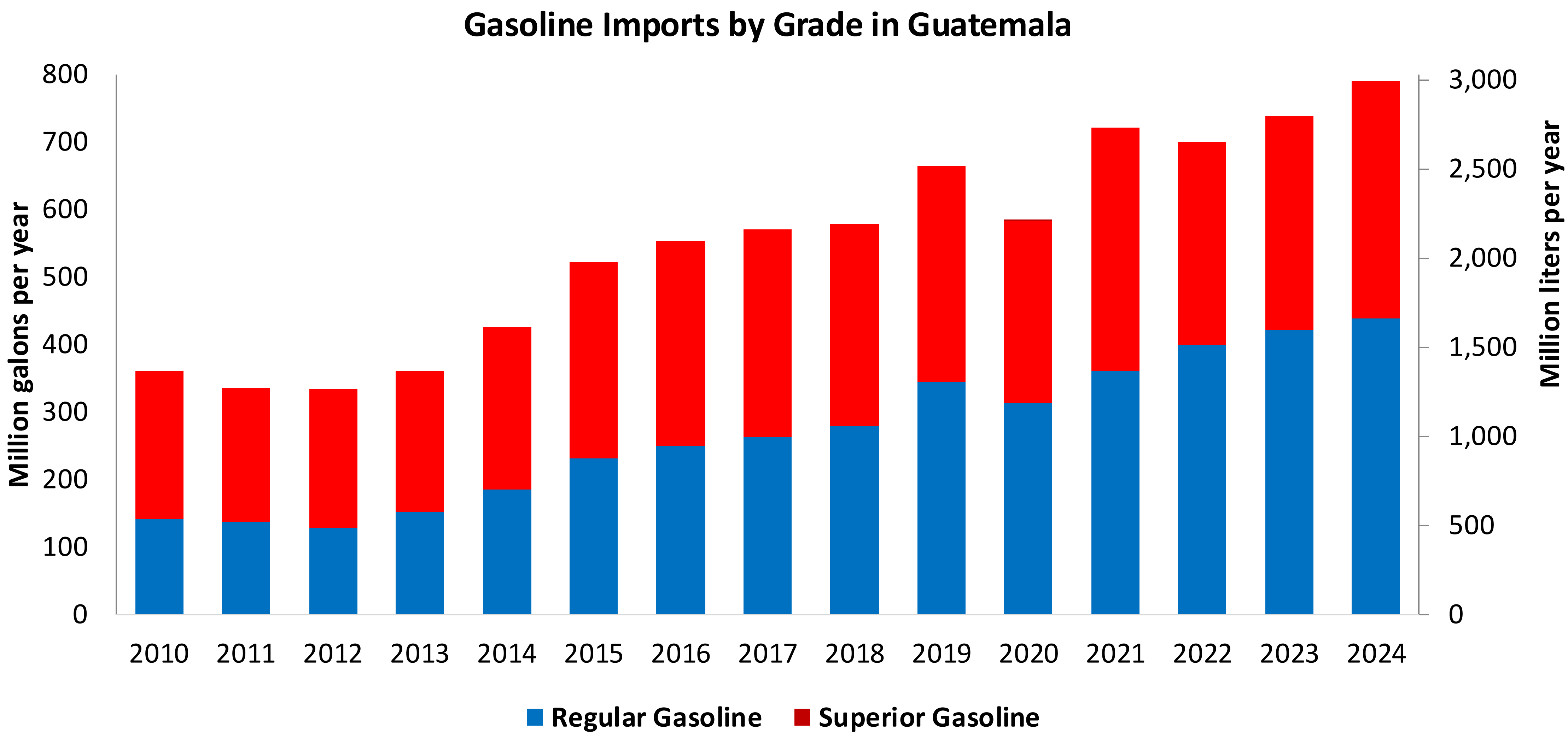
Source: MEM, 2025

Gasoline Demand in Guatemala



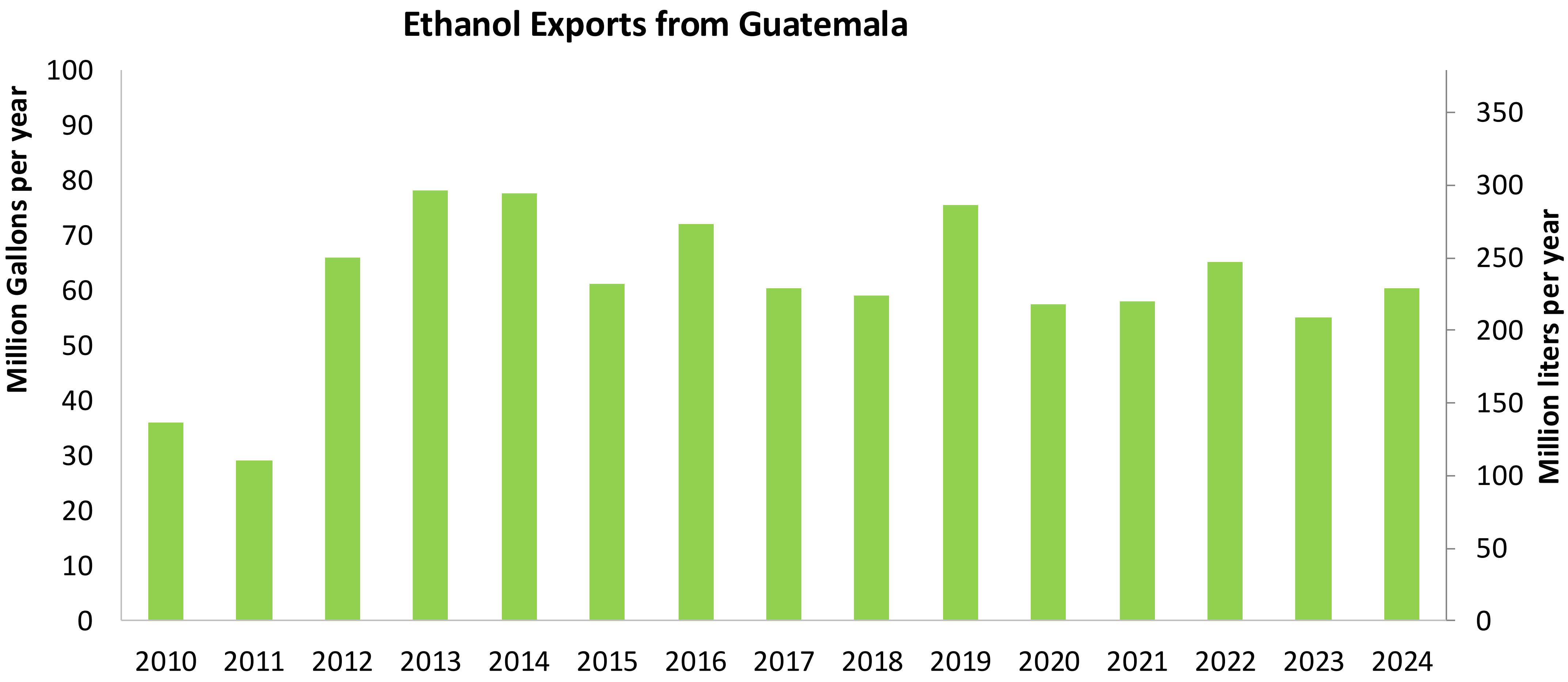
Source: MEM, 2025

Gasoline Imports in Guatemala



Source: MEM, 2025

Ethanol Exports from Guatemala



Source: Asociación de productores de etanol de Guatemala

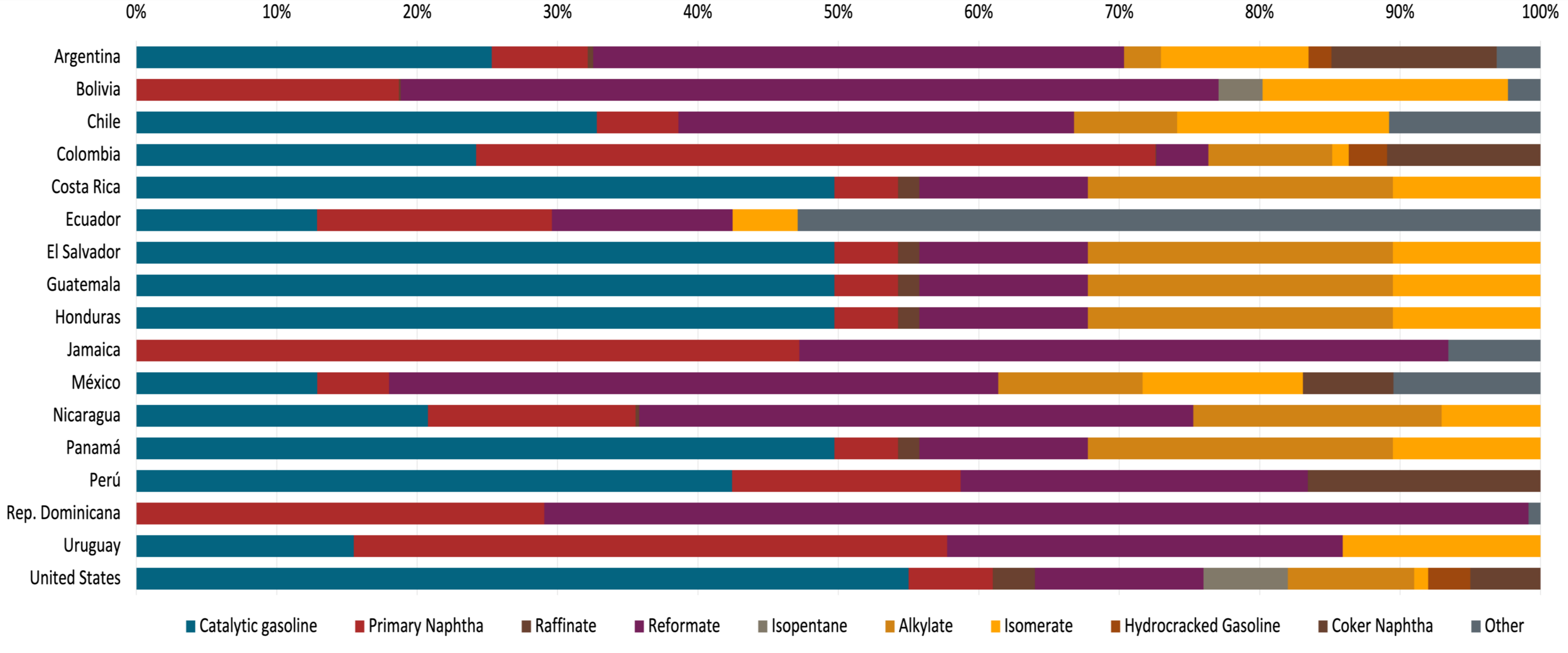
Gasoline Quality in Guatemala

Name	Ministerial Accord Number 320-2022		EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2022		2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	Gasoline Regular	Gasoline Superior	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 2,5% v/v	< 2,5% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	- / < 50% v/v	- / < 50% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	- / < 30% v/v	- / < 30% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 2,5 mg/l	< 2,5 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 91	> 95	> 95	> 95	> 98	> 98
MON	-	-	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 500 mg/kg	< 500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	-	-	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	< 10 %v/v	< 10 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 69 kPa	< 69 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	10% v/v	10% v/v	-	-	-	-
Ethers 5 or more C Atoms	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Source: MEM, 2022

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



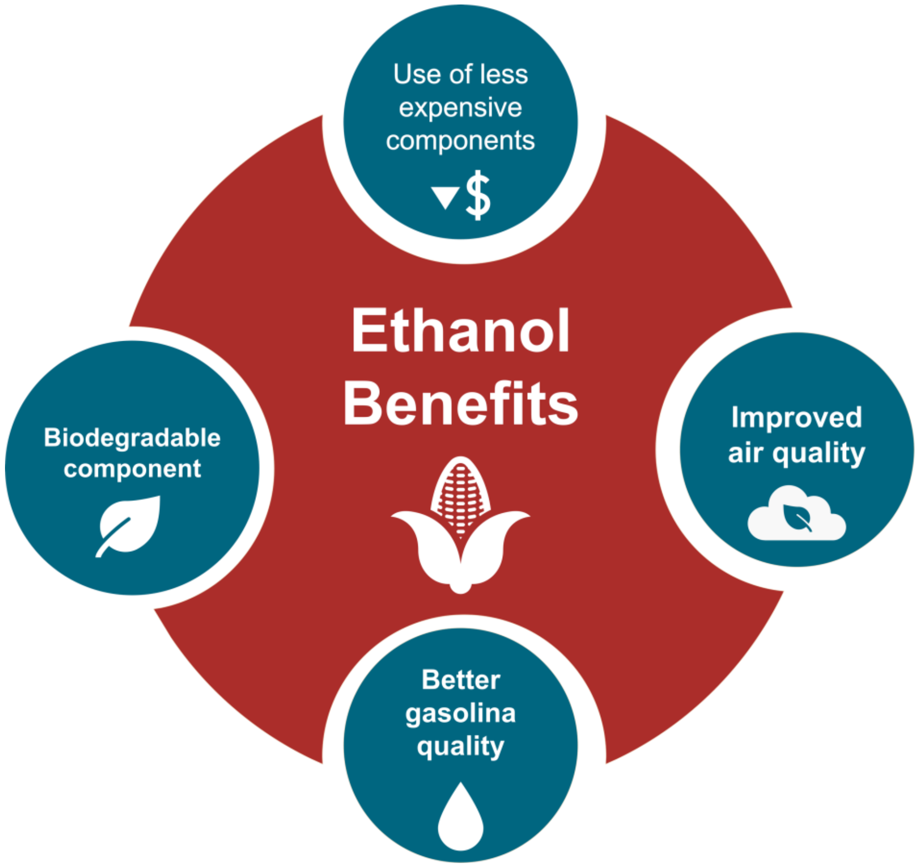
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

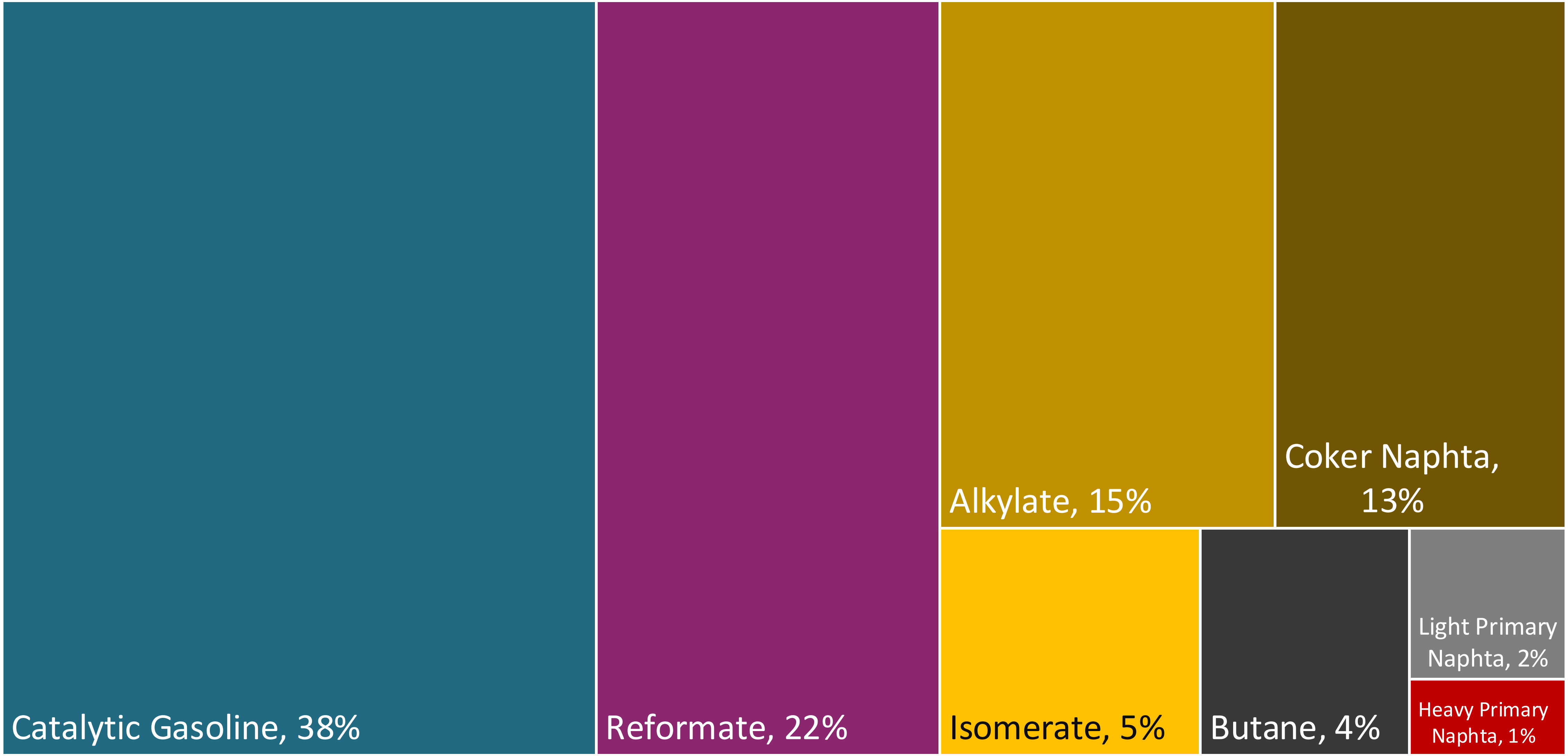
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average 2024 prices and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

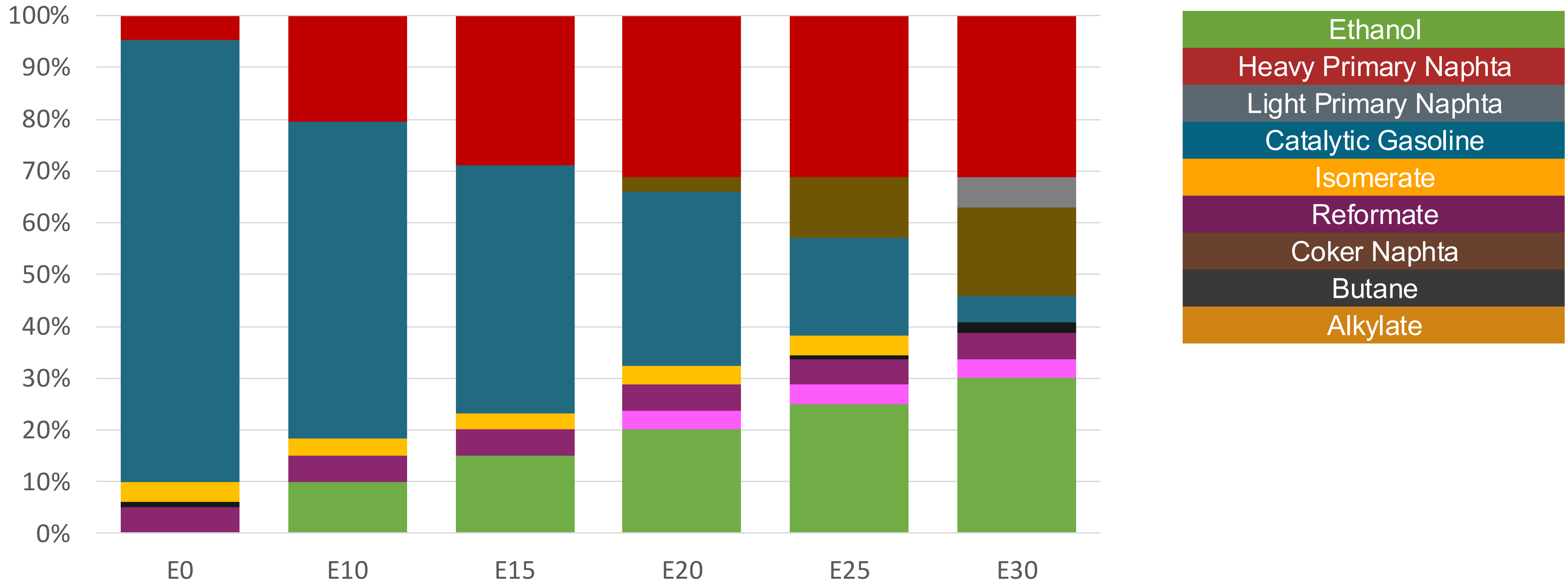


Blending Components - Guatemala



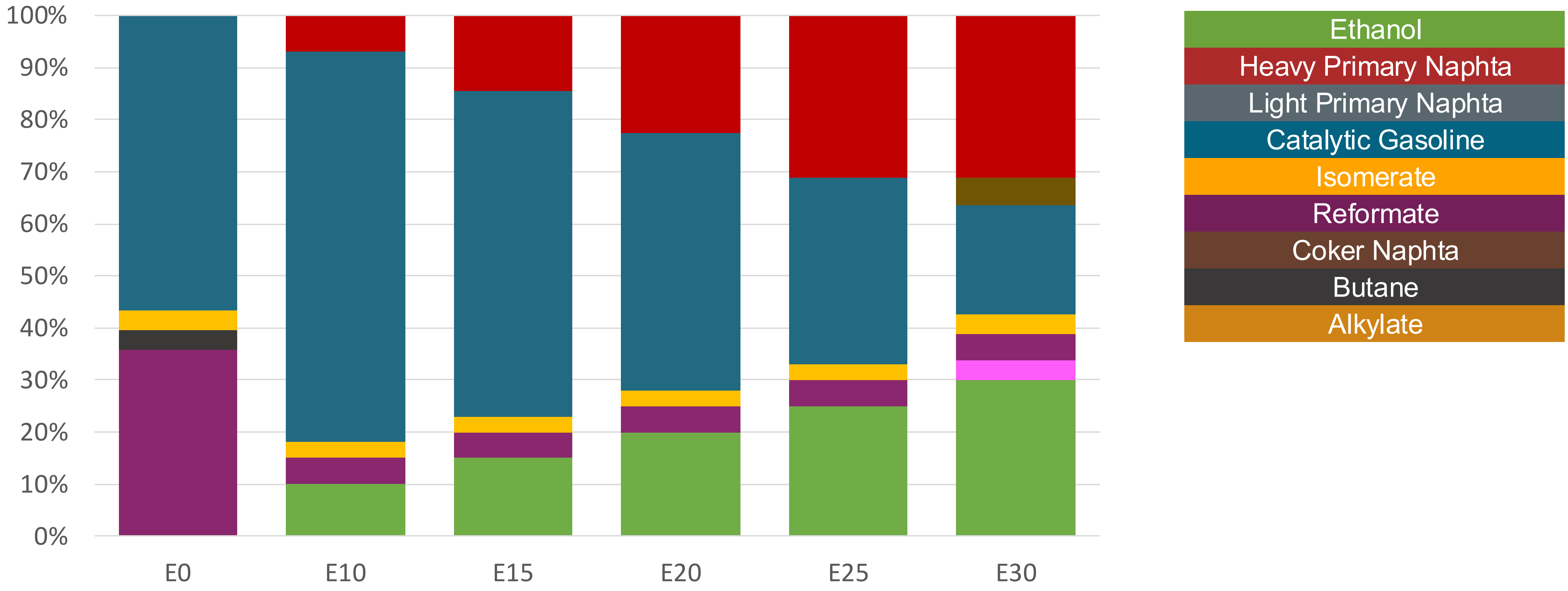
Source: Faro90

Guatemala – Regular – Constant Octane



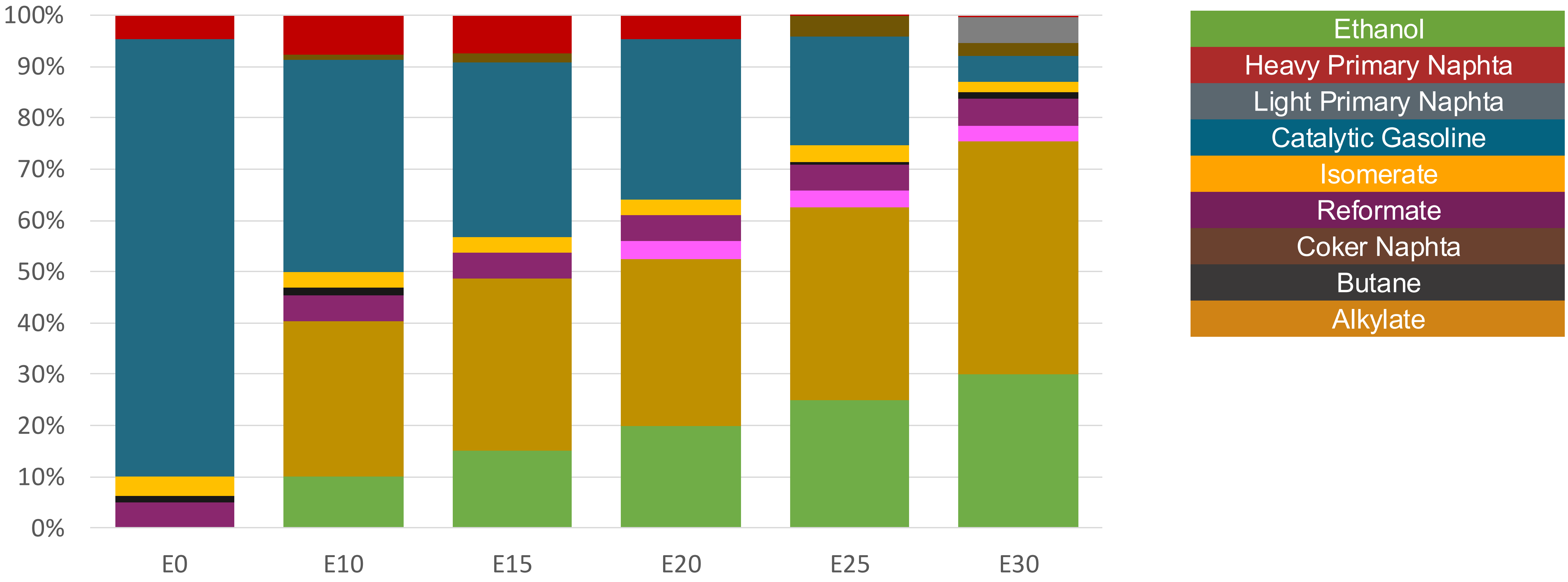
Octane (RON)	91.0	91.0	91.0	91.0	91.0	91.0
Price (USD/gal)	$ 2.336	$. 2.078	$. 1.934	$. 1.787	$. 1.636	$. 1.518

Guatemala – Premium – Constant Octane



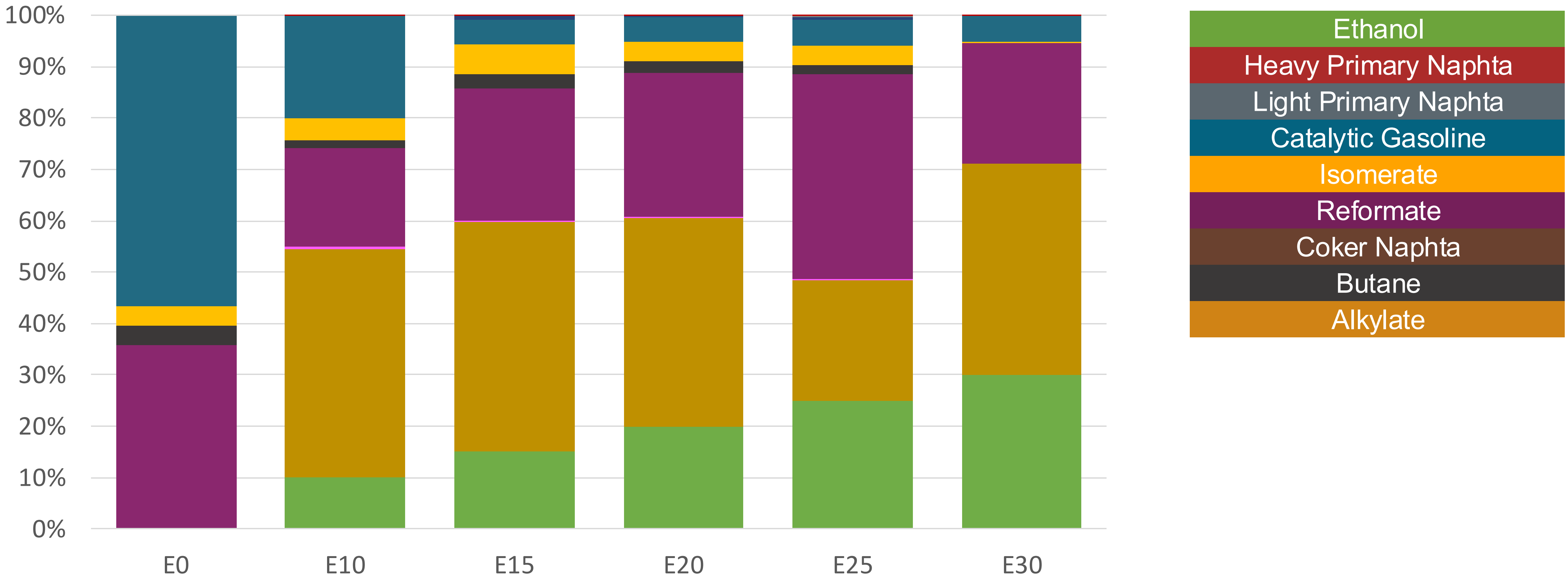
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (USD/gal)	$ 2.586	$ 2.283	$ 2.149	$ 2.007	$ 1.857	$ 1.709

Guatemala – Regular – Increased Octane



Octane (RON)	91.0	94.6	96.6	99.1	101.2	102.4
Price (USD/gal)	$ 2.336	$ 2.336	$ 2.336	$ 2.336	$ 2.336	$ 2.336

Guatemala – Premium – Increased Octane



Octane (RON)	95.0	98.0	100.3	102.6	104.9	105.8
Price (USD/gal)	$ 2.586	$ 2.586	$ 2.586	$ 2.586	$ 2.586	$ 2.586

Vehicle Emission Impact for Ethanol Gasoline Blending

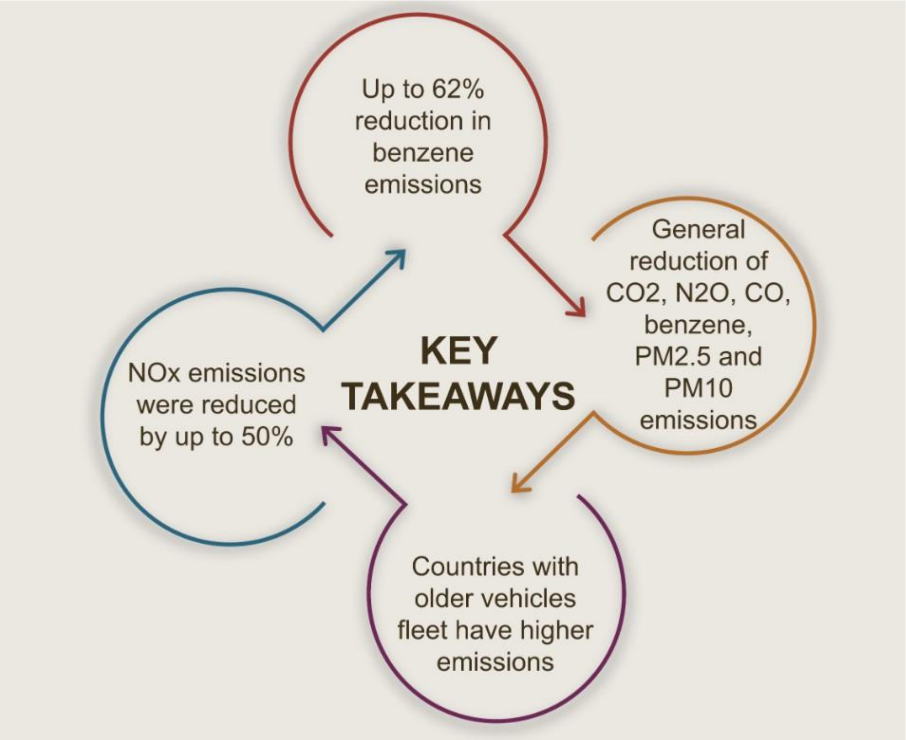
The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

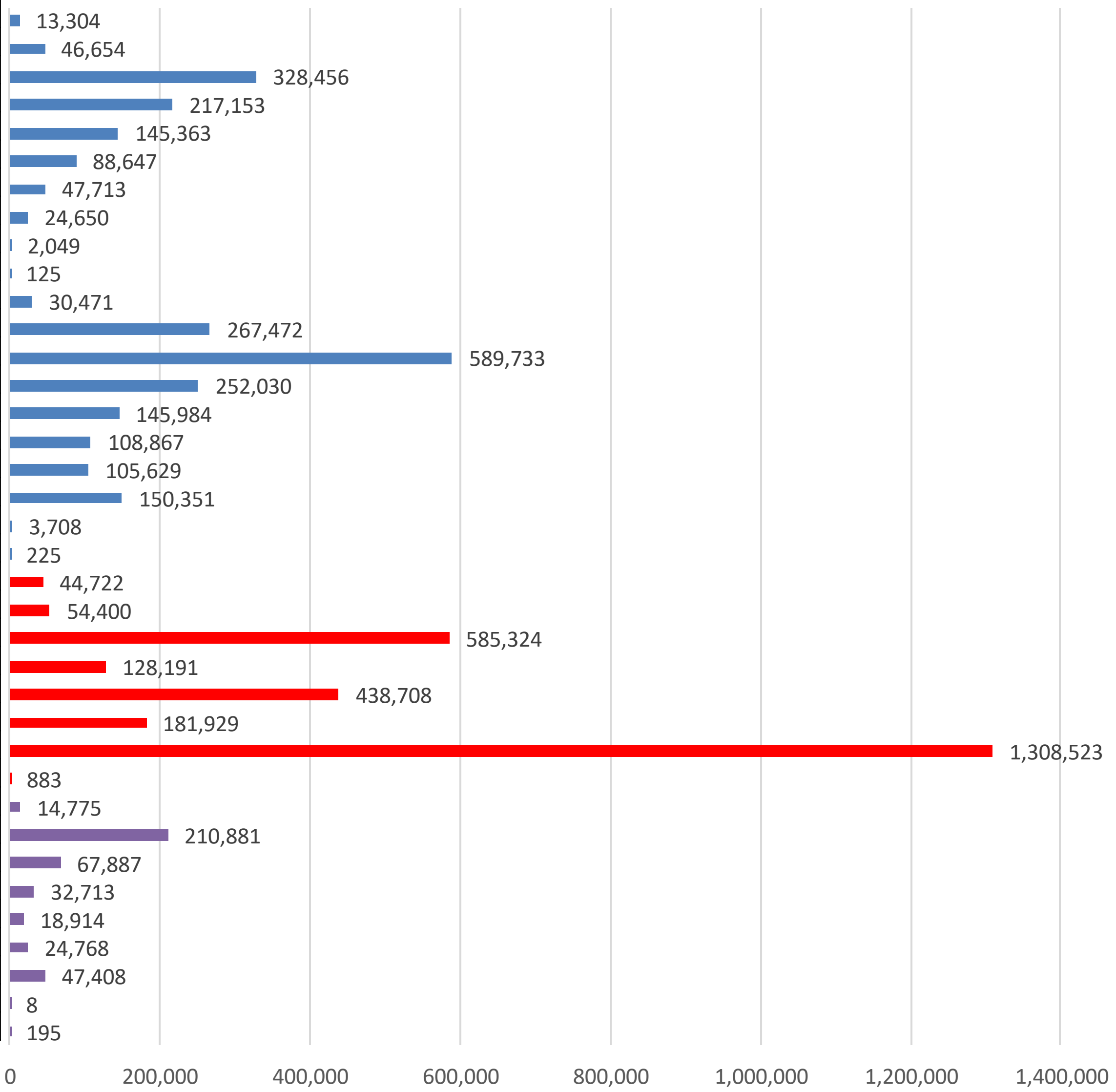
Main Results



**Source: Bureau of transportation statistics.*

Gasoline Vehicular Fleet - Guatemala

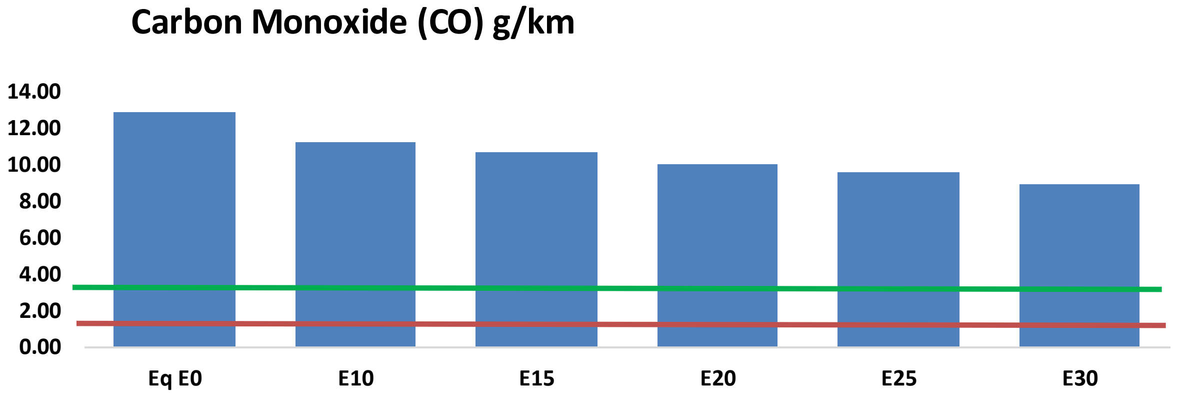
Type	Motor	Exhaust	Recuperator	Age	Number of Vehicles
Auto/Light	Carburator	3-Ways	PCV Tank	<8 years, < 160 mkm	13,304
Auto/Light	Single FI-Pt	3-Ways	PCV Tank	<8 years, < 160 mkm	46,654
Auto/Light	Multi FI-Pt	3-Ways	PCV Tank	<8 years, < 160 mkm	328,456
Auto/Light	Multi FI-Pt	Euro III	PCV Tank	<8 years, < 160 mkm	217,153
Auto/Light	Multi FI-Pt	Euro IV	PCV Tank	<8 years, < 160 mkm	145,363
Auto/Light	Multi FI-Pt	Euro V / Euro 6	PCV Tank	<8 years, < 160 mkm	88,647
Auto/Light	Multi FI-Pt	Euro V / Euro 6	PCV Tank	4-8 years, 80-160 mkm	47,713
Auto/Light	Multi FI-Pt	Euro V / Euro 6	PCV Tank	>4years, >80 mkm	24,650
Auto/Light	Hybrid			All	2,049
Auto/Light	Electric			All	125
Auto/Medium	Carburator	3-Ways	PCV Tank	<8 years, < 160 mkm	30,471
Auto/Medium	Single FI-Pt	3-Ways	PCV Tank	<8 years, < 160 mkm	267,472
Auto/Medium	Multi FI-Pt	3-Ways	PCV Tank	<8 years, < 160 mkm	589,733
Auto/Medium	Multi FI-Pt	Euro III	PCV Tank	<8 years, < 160 mkm	252,030
Auto/Medium	Multi FI-Pt	Euro IV	PCV Tank	<8 years, < 160 mkm	145,984
Auto/Medium	Multi FI-Pt	Euro V / Euro 6	PCV Tank	<8 years, < 160 mkm	108,867
Auto/Medium	Multi FI-Pt	Euro V / Euro 6	PCV Tank	4-8 years, 80-160 mkm	105,629
Auto/Medium	Multi FI-Pt	Euro V / Euro 6	PCV Tank	>4years, >80 mkm	150,351
Auto/Medium	Hybrid			All	3,708
Auto/Medium	Electric			All	225
Small Engines	FI 4-ciclos	None	PCV	<8 years, < 160 mkm	44,722
Small Engines	FI 4-ciclos	Euro II	PCV	<8 years, < 160 mkm	54,400
Small Engines	FI 4-ciclos	Euro III	PCV	<8 years, < 160 mkm	585,324
Small Engines	FI 4-ciclos	Euro IV	PCV	<8 years, < 160 mkm	128,191
Small Engines	FI 4-ciclos	Euro IV	PCV	4-8 years, 80-160 mkm	438,708
Small Engines	FI 4-ciclos	Euro V	PCV	4-8 years, 80-160 mkm	181,929
Small Engines	FI 4-ciclos	Euro V	PCV	>4years, >80 mkm	1,308,523
Small Engines	Eléctricos			All	883
Trucks	Carburator	3-Ways	PCV	<8 years, < 160 mkm	14,775
Trucks	FI	3-Ways	PCV	<8 years, < 160 mkm	210,881
Trucks	FI	Euro III	PCV	<8 years, < 160 mkm	67,887
Trucks	FI	Euro IV	PCV	<8 years, < 160 mkm	32,713
Trucks	FI	Euro V / Euro 6	PCV	<8 years, < 160 mkm	18,914
Trucks	FI	Euro V / Euro 6	PCV	4-8 years, 80-160 mkm	24,768
Trucks	FI	Euro V / Euro 6	PCV	>4years, >80 mkm	47,408
Trucks	Hybrid			All	8
Trucks	Electric			All	195

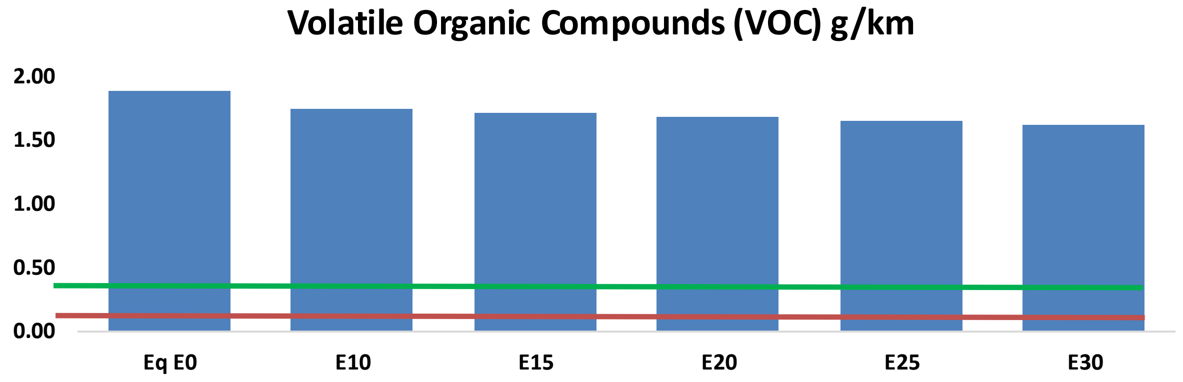


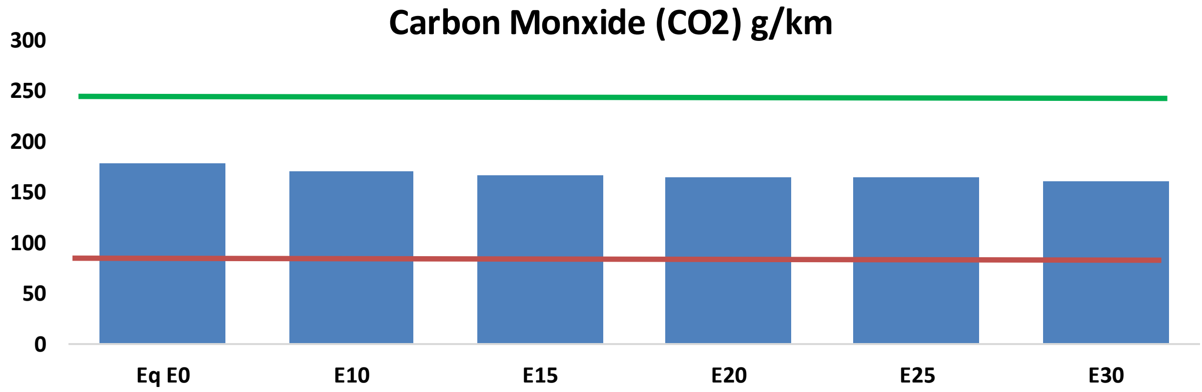
Gasoline Vehicular Fleet: **5,728,610**
Average age: **13 years**
Motorcycles: **48%**
Source: INE, analysis Faro 90

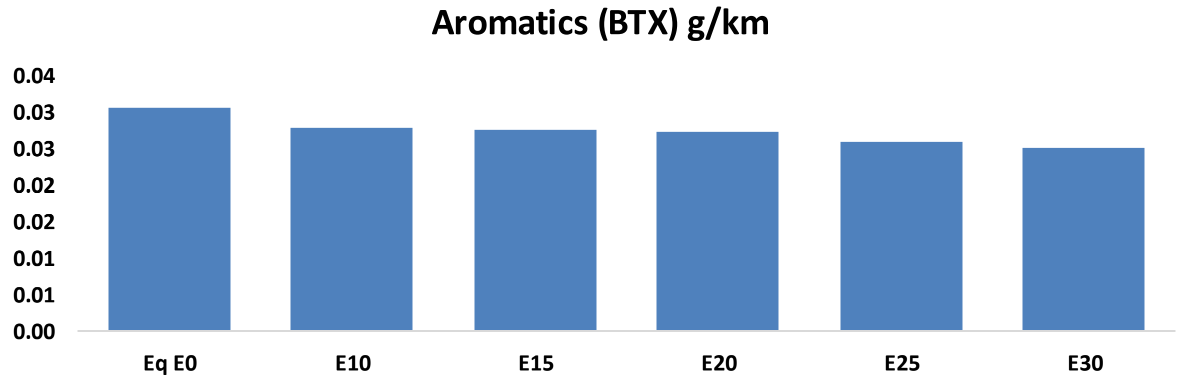
Guatemala – Vehicular Emissions

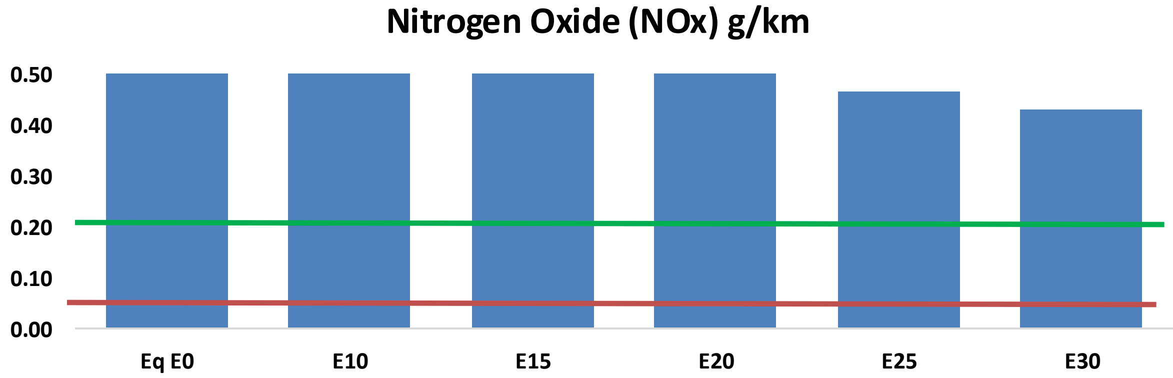
United States
Euro 6

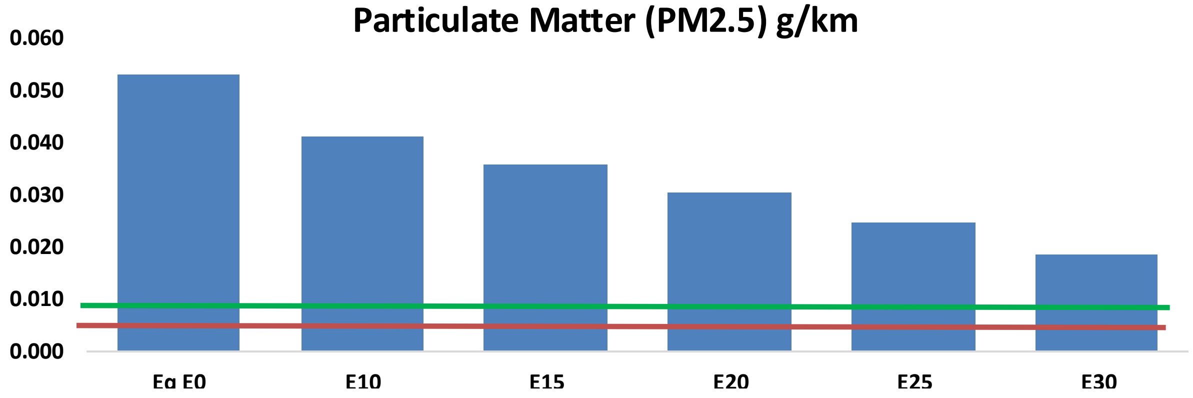












Guatemala – Vehicular Emissions

Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,248,309	1,170,157	1,092,005	1,031,438	972,088	926,933	870,637	-13%	-22%
CO2	17,319,676	16,874,313	16,453,692	16,122,886	15,959,191	15,915,024	15,621,664	-5%	-8%
VOC	182,094	175,747	169,400	165,691	162,647	160,513	156,933	-7%	-11%
NOx	77,928	58,446	54,550	51,413	48,487	45,261	41,848	-30%	-38%
SOx	213	197	181	167	154	140	125	-15%	-28%
NH3	6,841	6,773	6,705	6,737	6,813	6,866	6,910	-2%	0%
N2O	317	315	314	315	322	325	329	-1%	2%
CH4	40,547	40,547	40,547	41,156	42,048	42,818	43,548	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	697	653	610	576	543	517	483	-13%	-22%
Acetaldehyde	1,717	1,971	2,892	4,404	6,000	6,857	8,102	68%	249%
Formaldehyde	6,428	6,250	7,285	8,555	8,962	9,915	10,793	13%	39%
Benzene	2,977	2,861	2,711	2,668	2,645	2,530	2,448	-9%	-11%
PM2.5	1,076	956	835	725	616	500	379	-22%	-43%
PM10	4,030	3,578	3,126	2,715	2,306	1,871	1,418	-22%	-43%

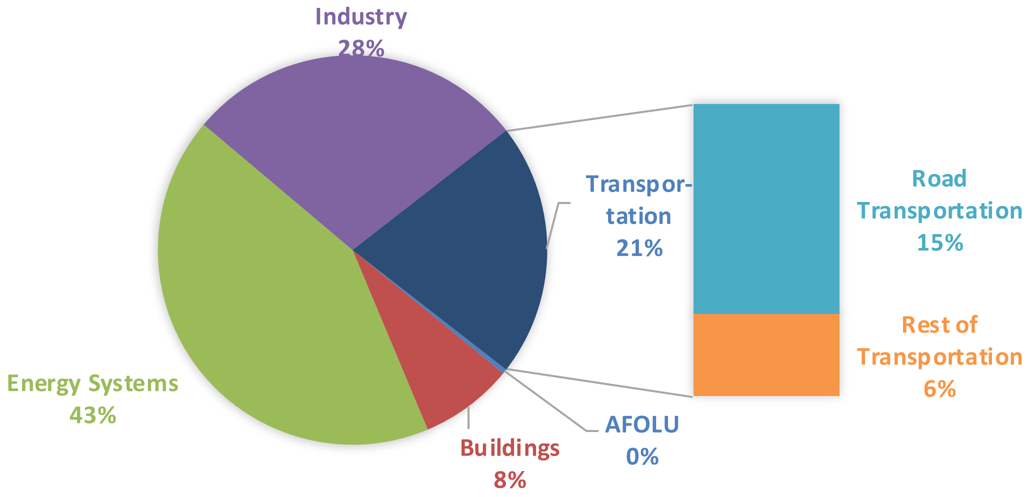
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

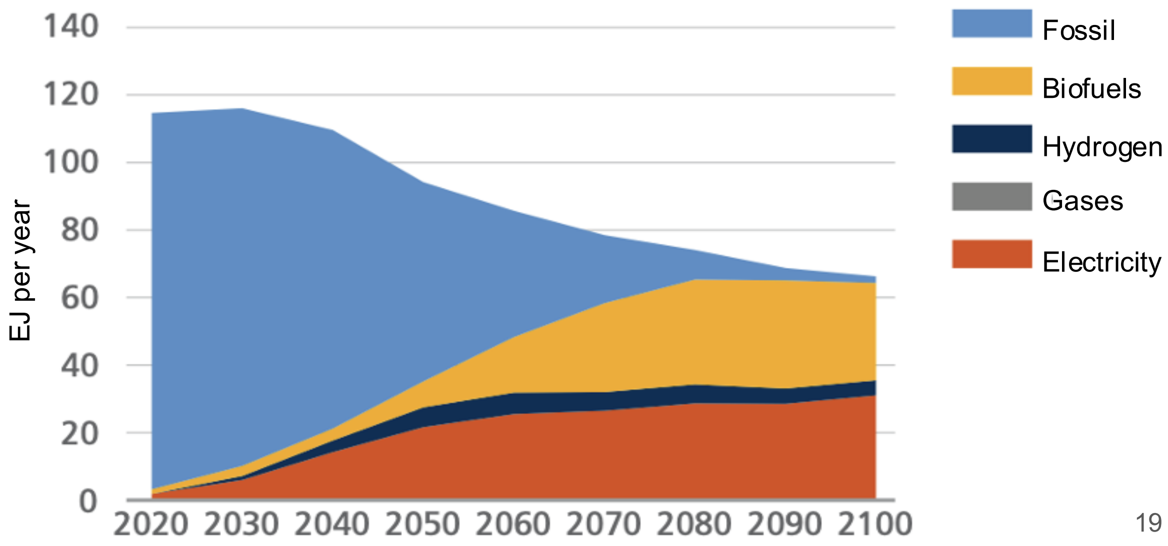
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

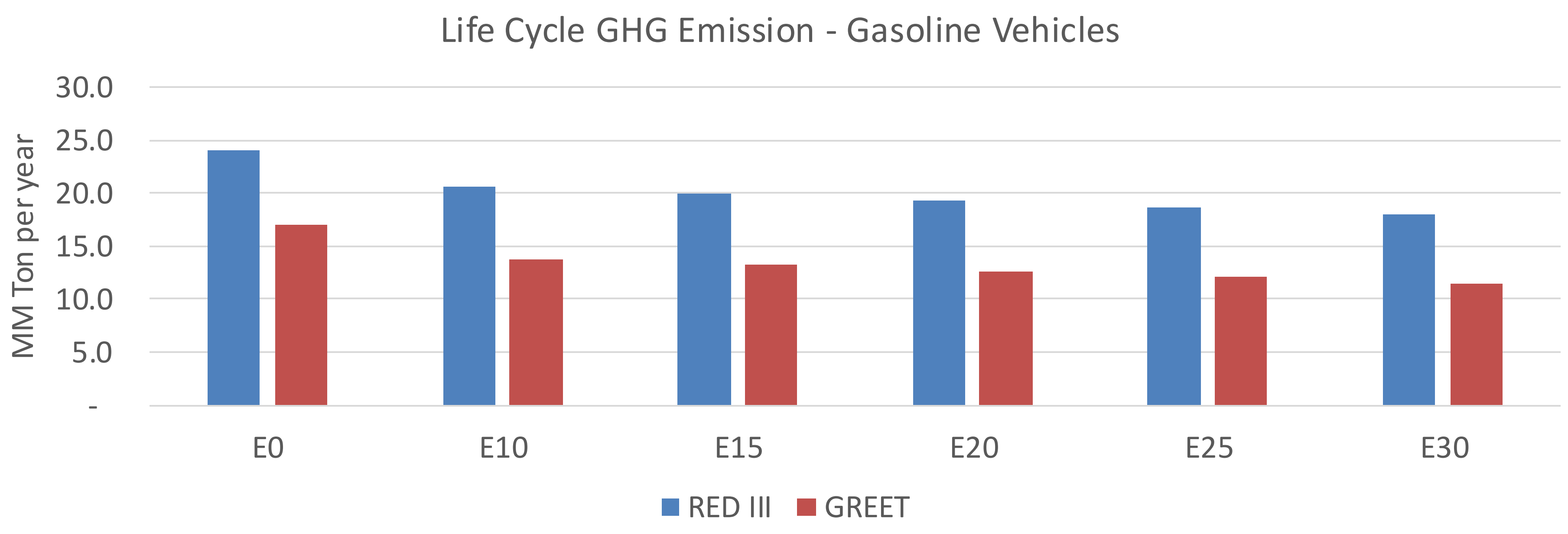
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Guatemala – Gasoline Vehicle Fleet GEI Emissions



Emisiones País 2024 (MMT/año)	43.5
Meta a 2035 (MMT/año)	21.7
Delta	21.7

%Reducción vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	18.6%	22.2%	25.3%	28.4%	32.2%
RED II/III	0.0%	14.0%	17.0%	19.5%	22.1%	25.1%

% Meta@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	14.5%	17.3%	19.7%	22.2%	25.1%
RED II/III	0.0%	15.5%	18.7%	21.6%	24.4%	27.8%

Source: Greet, RED II/III, climateactiontracker.org, F90